

Ganesh Gupta

817-822-1131 | iamgs10rk@gmail.com | linkedin.com/in/ganeshhgupta | github.com/ganeshhgupta

EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Wrote infrastructure-level **Go** and **Rust** scheduling and **fault-tolerant placement** logic for a multi-cloud **GPU/HPC fleet** (AWS, GCP, CoreWeave), reasoning about failure modes and blast radius under capacity-scarcity constraints.
- Engineered a **distributed executor** orchestrating pipeline-node lifecycle (creation, scale-up, failover) across **Docker** and serverless compute, via typed **REST APIs** at **sub-200ms p99 latency**.
- Operated **NVIDIA GPU clusters** with **InfiniBand/RoCE** networking, maintaining fleet utilization through node loss and provider instability via **Slurm-based HPC scheduling**.
- Implemented **OAuth**-based auth and **RBAC** for multi-tenant **isolation**; used **AI coding assistants** (Claude Code, Codex) across design, implementation, and review, validating generated code against correctness and security bars.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Rebuilt a synchronous service into an **async, autoscaled** architecture on **GCP Cloud Run**, cutting **p99 latency from 420ms to 140ms** across **50K+ daily requests**.
- Triageed and resolved **distributed-system incidents** end-to-end on an **on-call rotation**, driving **root cause analysis** and operational improvements with automated **CI/CD rollback**.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Architected and operated **25+ production Java microservices** for **1M+ users** at **99.9% uptime** over 4 years on **AWS (EC2, RDS, SQS)**, owning **system design** end-to-end.
- Standardized **retry, timeout, and failure-propagation** contracts across 25+ services, cutting production escalations **60%** in one quarter via **root cause analysis** and post-incident reviews.
- Led **architecture and code-review** sessions, consolidating duplicate service abstractions and mentoring engineers on **Object-Oriented Design** and reliability.
- Cut a nightly SQL aggregation pipeline from **8 minutes to 90 seconds** via **indexed views** and batch partitioning, eliminating row-level lock contention.

PROJECTS

Distributed KV Store

github.com/ganeshhgupta/Distributed-KV-Store

- Built a replicated key-value store in **Go** implementing **Raft consensus**, **linearizable reads**, and **log compaction**, tolerating node failures without data loss.

VPC Control Plane Simulator

github.com/ganeshhgupta/vpc-control-plane-sim

- Built an AWS-style **control-plane** simulator with **CIDR** validation, **longest-prefix-match routing**, stateful security-group evaluation, and configuration **drift detection**.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Built a self-hosted observability platform tracing requests, **latency**, and cost through a **FastAPI** collector and **PostgreSQL** store, with a **React** dashboard.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Aug 2016 – Jul 2020

SKILLS

Languages: Go, Rust, Java, Python, C++, SQL

Distributed Systems: Control-Plane Orchestration, Raft Consensus, Fault-Tolerant Placement, Multi-Tenant Isolation, Object-Oriented Design

Cloud & Infra: AWS, GCP, Docker, CI/CD, Slurm-based HPC Scheduling

Engineering: Design Reviews, Code Review, On-Call, Root Cause Analysis, AI-Assisted Development

Certifications: AWS Certified Developer Associate, GCP Professional MLOps Engineer